Mohammed Aayan Pathan

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Work and Volunteering Experience

Junior Engineering Associate — Full-Stack Web Developer Oct 2025 — Apr 2026
VOXA (voxa.club) *Dubai, UAE (Remote)*

- Refactored the Voxa MVP codebase into production-ready version 1.0, modernizing architecture with redefined user stories and responsive design.
- Co-developed investor-facing Demo app deployed at Lisbon WebSummit 2025 using React (Vite) and Node.js; responsible for version control and deployment workflow.
- Optimized UX and UI design alongside colleagues, providing assistance and insight with Figma designs to reduce navigation friction.
- Implemented Firebase Functions + App Check to automatically flag and remove 100% of harmful community content using AI-based moderation in the MVP.
- Reworked Google Analytics integrations, enabling more precise tracking of user-interaction data for data-driven UX iterations.
- Assisted in onboarding 10+ developer interns: authored a 15+ page developer wiki, provided code reviews, and enforced Git workflow standards.
- Lead migration by transitioning v1.0 authentication and data layer from Firebase to Supabase (PostgreSQL) to reduce technical debt and improve schema scalability.

Product and App Development Intern July 2025
VOXA (voxa.club) *Dubai, UAE (Remote)*

- Built a multi-step onboarding flow with Firebase Auth (email, Google, and GitHub sign-in), proactively initialized a structured user object from OAuth data prior to account creation to enable personalized state from first login.
- Developed a card-based practice system integrating AssemblyAI transcription with OpenAI GPT feedback generation; each session injected the transcription alongside card metadata into a structured prompt returning categorized feedback including filler word detection.
- Built a community forum with Firestore-backed posts, replies, image and recording attachments, upvoting, bookmarking, and real-time moderation flagging on submission
- Conducted user testing with 30+ external testers to gather feedback for iterative improvements.

GCSE Computer Science Intervention Volunteer Tutor Sept 2024 — Apr 2025
GEMS Founders School - Al Barsha *Dubai, UAE*

- Co-taught twice-weekly GCSE Intervention sessions alongside the Head of Computer Science for Year 11 and Year 12 students.
- Marked mock programming, algorithms, and theory examinations; directly contributed to improved student grades (failing grades to passing, passing grades to A grades) and increased A-Level CS enrollment.
- Awarded Certificate of Appreciation in Computer Science — sole Year 12 recipient in the cohort.

Engineering Projects

Full-Stack Developer, FormulaMetric (formulametric.com) May 2025 — Present

- Designed full-stack F1 analytics platform using React, Node.js, Express, and MySQL with normalized schema, foreign keys, and indexed lookups.
- Created RESTful API with season, driver, and race filters, including pagination and query parameters.
- Implemented data ingestion from the Jolpica API with a recursive exponential backoff retry algorithm to handle rate limits.
- Built regression, correlation, and sigmoid-based performance metrics for multi-season driver comparisons.
- Developed driver and team assignment logic including support for rotating lineups across seasons.
- Utilized a random-pivot recursive Quicksort algorithm to sort drivers by any of the available metric factors.
- Designed a top-k min-heap (binary tree with priority queue operations) to efficiently find the top 3 drivers in a season in linear time.

- Implemented loading and fallback states to ensure smooth UX across data-intensive views.
- Built interactive UI with MUI X gauges, charts, and responsive layouts supporting comparison of up to 3 drivers.

Developer, UniMetric (syntaxsnipes.github.io/UniMetric) June 2026

- Built a client-side university ranking with approximately 1500 university institutions across 5 years of Quacquarelli Symonds (QS) World University Ranking data, parsed and normalized from raw excel exports into JSON.
- Implemented a proportional weight normalization algorithm that preserves the relative weighting proportions from the original rankings to accommodate a selection of different factor metrics, ensuring consistent re-ranking regardless of selected combination
- Developed per-metric trend charts across 5 years for each university, showing score and rank trajectories that the normal QS rankings data doesn't show.
- Built a head-to-head comparison mode supporting up to 3 universities simultaneously across all metrics with interactive shadcn/ui charts.
- Implemented shareable URL encoding of selected filters, year and country for persistent and distributable custom rankings.

Team Leader, Full-Stack Developer, TLSearch, NASA Space Apps Challenge (tlsearch.org) Oct 2025 — Mar 2026

- Built a semantic search engine for NASA space-biology datasets using vector embeddings and cosine similarity ranking.
- Implemented FastAPI backend computing in-memory similarity scores and returning ranked JSON results.
- Loaded NASA datasets into Python and processed them in-memory for fast retrieval and experimentation.
- Developed Next.js frontend with TailwindCSS and SEO-ready metadata connected to the FastAPI backend.

Education

American University of Sharjah Aug 2026 — May 2030
BSc Computer Science *Sharjah, UAE*

- Awarded highest Merit Scholarship (20%), granted to applicants with 95%+ average

GEMS Founders School - Al Barsha Aug 2022 — July 2026
A-Levels (GCE/IAL) *Dubai, UAE*

- **Achieved:** A*A*AA
- **Subjects:** Mathematics, Computer Science, Further Mathematics, Physics

GEMS Founders School - Al Barsha Aug 2020 — June 2024
GCSE / iGCSE *Dubai, UAE*

- **Achieved:** 999988877 (A*A*A*A*A*A*A*AA)
- **Subjects:** Mathematics, Further Pure Mathematics, Computer Science, Physics, Chemistry, Psychology, Business Studies, and English Literature/Language

Skills

- **Languages:** TypeScript, JavaScript, Python, SQL (MySQL, PostgreSQL), HTML, CSS
- **Frameworks & Libraries:** Next.js, React, Node.js, Express.js, FastAPI, TailwindCSS, shadcn/ui
- **Cloud & Databases:** Supabase, Firebase, MySQL, PostgreSQL, Vercel, Render, Aiven
- **Tools & Methodologies:** Git, GitHub, Agile Sprints, Figma, WebSockets, REST APIs

Awards, Honors and Certificates

- Star Associate of the Year — VOXA
- ICSC Bronze Finalist w/honors (top 8% internationally)
- Best Student — Year 12 Computer Science Cohort (2024–2025)
- 2× UKMT Intermediate Mathematical Challenge Bronze
- Claude Code in Action, Claude Code 101, AI Fluency of Builders — Anthropic Academy